

JNTUK R23 Regulation | Semester - I

BASIC ELECTRICAL & ELECTRONICS ENGINEERING (BEEE)

UNIT - III

Energy Resources, Electricity Bill & Safety Measures

Prepared for B.Tech First Year Students

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Table of Contents

1. Introduction to Energy Resources	3
2. Energy Resources Overview	3
3. Hydel Power Plant	4
4. Nuclear Power Plant	6
5. Solar Energy System	8
6. Wind Energy System	9
7. Comparison of Energy Sources	11
8. Electricity Bill Concepts	12
9. Power Rating Table of Appliances	13
10. Numerical Problems on Electricity Bill	13
11. Fuse - Working, Merits, Demerits	15
12. MCB - Working, Merits, Demerits	16
13. Earthing, Electric Shock & Safety Measures	17
14. Quick Revision Sheet	20

Introduction to Energy Resources

Energy is the fundamental requirement for the development of any nation. **Electrical energy** is the most versatile form of energy because it can be easily transported, controlled, and converted to other forms. The entire modern industrial, domestic, and commercial infrastructure relies on a continuous and reliable supply of electrical energy.

Energy resources are broadly classified into two categories:

Conventional (Non-Renewable) Energy Sources: These are traditional sources that have been in use for a long time. Examples: Thermal, Hydel, Nuclear power plants. They are depleting and not eco-friendly (except hydel/nuclear).

Non-Conventional (Renewable) Energy Sources: These are inexhaustible and eco-friendly sources available in nature. Examples: Solar, Wind, Tidal, Geothermal, Biomass energy.

The selection of energy source depends on factors like availability of raw material, geographical location, load requirements, initial cost, running cost, and environmental impact. This unit covers all major sources along with their working principles, layouts, and diagrams as prescribed in the JNTUK R23 syllabus.

Energy Resources Overview

2.1 Conventional Energy Sources

A. Thermal Power Plant

A **Thermal Power Plant** generates electricity by burning fossil fuels like coal, oil, or natural gas to produce steam, which drives a turbine connected to a generator. It is the most widely used source of power in India.

Working: Fuel → Boiler → Steam → Turbine → Generator → Electricity

✓ Advantages

- Can be installed anywhere
- Less initial cost than hydel
- Can generate power as per demand
- Reliable and continuous generation

✗ Disadvantages

- Causes air and water pollution
- Depletes fossil fuels (non-renewable)
- High running cost
- Produces greenhouse gases (CO₂, SO₂)

B. Overview of Energy Classification

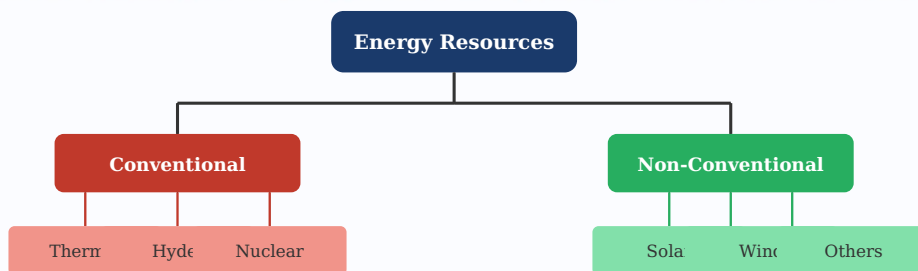


FIG 2.1 - CLASSIFICATION OF ENERGY RESOURCES

Hydel Power Plant (Hydroelectric Power Plant)

3.1 Definition

A **Hydel (Hydroelectric) Power Plant** converts the potential energy of stored water at a height into electrical energy. It is one of the oldest and cleanest forms of energy generation.

3.2 Principle of Operation

Working Principle: The potential energy of water stored at a height is converted into kinetic energy as it flows down. This kinetic energy rotates the turbine blades. The turbine is coupled to a generator that converts mechanical energy into electrical energy.

KEY FORMULAS

Power (P) = $\rho \times g \times Q \times H \times \eta$ (Watts)

where: ρ = density of water (1000 kg/m³), g = 9.81 m/s², Q = flow rate (m³/s), H = head (m), η = efficiency

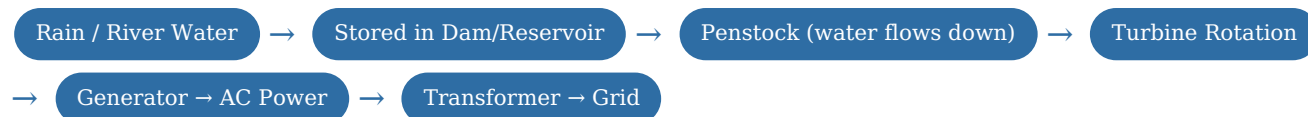
Energy Generated = $P \times t$ (kWh)

3.3 Construction & Layout

A hydel power plant consists of the following main components:

1. **Dam / Reservoir:** Stores large volumes of water at a considerable height to create potential energy.
2. **Penstock:** Large pipe or tunnel that carries water from the reservoir to the turbine.
3. **Surge Tank:** Relieves excess pressure in penstock during load variations.
4. **Water Turbine (Pelton/Francis/Kaplan):** Converts water's kinetic energy to mechanical (rotational) energy.
5. **Generator:** Converts mechanical energy from turbine into electrical energy (AC).
6. **Transformer:** Steps up the voltage for long-distance transmission.
7. **Tailrace:** Channel that carries water away from the turbine after generating power.

3.4 Working Steps (Step-by-Step)



3.5 Neat Labeled Diagram

HYDEL POWER PLANT - LAYOUT DIAGRAM

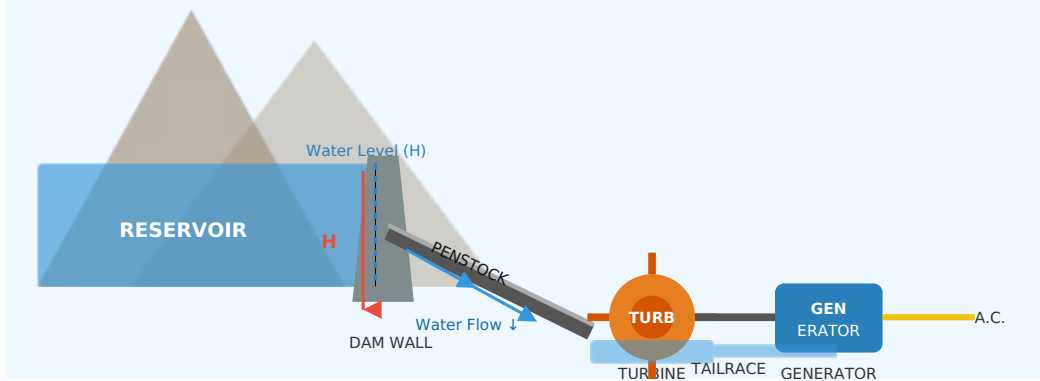


FIG 3.1 - HYDEL POWER PLANT LAYOUT

✓ Advantages

- Uses renewable water resource
- No air pollution / fuel cost
- Very long life (50-100 years)
- Low operating & maintenance cost
- Quick start/stop capability

✗ Disadvantages

- Depends on rainfall & geography
- Very high initial capital cost
- Large area submerged (ecological loss)
- Affects aquatic ecosystem & community
- Silting reduces capacity over years

3.6 Applications

- Base load and peak load power stations
- Irrigation and flood control (multi-purpose dams)
- Examples: Bhakra Nangal (India), Three Gorges (China), Hoover Dam (USA)

Nuclear Power Plant

4.1 Definition

A **Nuclear Power Plant** generates electricity using heat energy released from the **nuclear fission** of heavy atoms like Uranium-235 (U-235) or Plutonium-239 (Pu-239). The heat converts water into steam, which then drives a turbine-generator set.

4.2 Principle - Nuclear Fission

Nuclear Fission: When a slow neutron strikes the nucleus of U-235, it splits into two smaller nuclei (Barium + Krypton), releasing 2-3 neutrons and enormous amounts of heat energy. These neutrons trigger a chain reaction, continuously producing heat.

EINSTEIN'S MASS-ENERGY EQUIVALENCE

$$E = mc^2$$

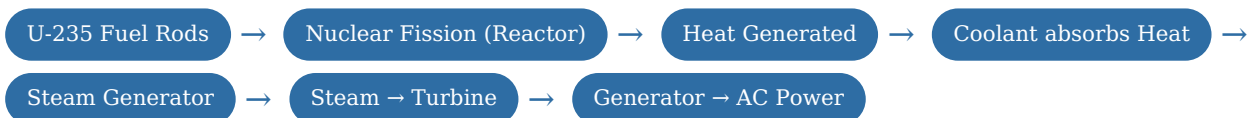
where: E = energy (Joules), m = mass defect (kg), c = speed of light (3×10^8 m/s)

1 gram of U-235 releases $\approx 8 \times 10^{10}$ Joules of energy

4.3 Main Components

1. **Reactor Core:** Contains nuclear fuel (Uranium) rods in which fission takes place.
2. **Moderator:** Slows down fast neutrons to sustain the chain reaction (heavy water, graphite).
3. **Control Rods:** Made of Cadmium or Boron; absorb excess neutrons to control reaction rate.
4. **Coolant:** Carries heat from reactor to heat exchanger (water, heavy water, liquid sodium).
5. **Heat Exchanger / Steam Generator:** Transfers heat from coolant to produce steam.
6. **Turbine:** Steam drives the turbine which is coupled to the generator.
7. **Generator:** Converts mechanical energy into electrical energy.
8. **Shielding:** Thick concrete walls to prevent radiation leakage.

4.4 Working (Step-by-Step)



4.5 Neat Labeled Diagram

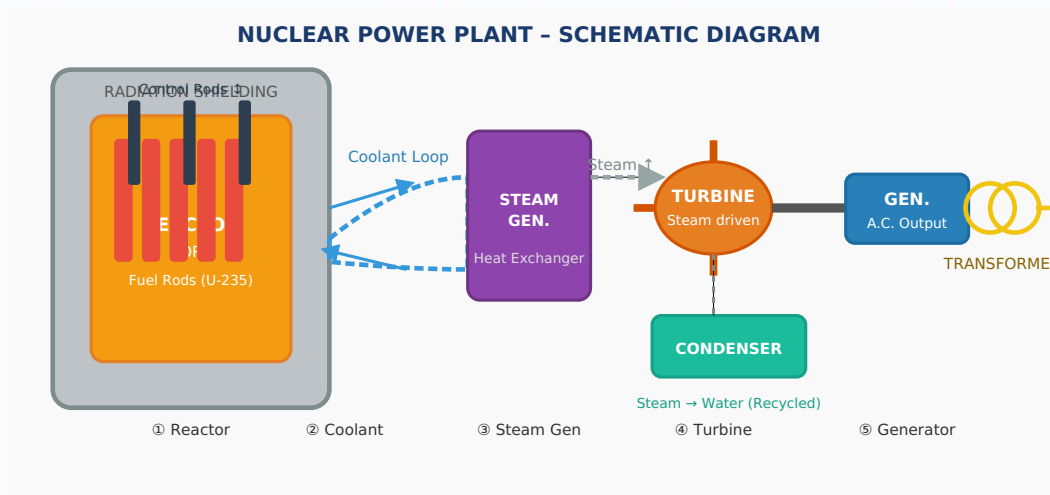


FIG 4.1 - NUCLEAR POWER PLANT SCHEMATIC

✓ Advantages

- Huge energy from very small fuel
- No greenhouse gas emission
- Low fuel cost; small storage needed
- Reliable baseload power generation

✗ Disadvantages

- High initial and construction cost
- Radioactive waste disposal problem
- Severe accident risk (Chernobyl, Fukushima)
- Requires highly trained personnel

Solar Energy System

5.1 Definition

A **Solar Power System** converts sunlight directly into electricity using **Photovoltaic (PV) cells**. When sunlight (photons) strikes the semiconductor material (Silicon), electron-hole pairs are generated, creating a voltage — this is the *Photovoltaic Effect*.

KEY FORMULAS

Solar Power Output: $P = G \times A \times \eta$

where: G = Solar irradiance (W/m^2), A = Panel area (m^2), η = Panel efficiency

Efficiency $\eta = (\text{Electrical Output Power}) / (\text{Incident Solar Power}) \times 100\%$

Standard irradiance = 1000 W/m^2 (at STC)

5.2 Components of a Solar Power System

1. **Solar Panels (PV Modules):** Array of PV cells that convert sunlight to DC electricity.
2. **Charge Controller:** Regulates voltage/current from panels to battery; prevents overcharging.
3. **Battery Bank:** Stores DC electrical energy for use during night or cloudy periods.
4. **Inverter:** Converts DC from panels/battery to AC for household/grid use.
5. **Grid / Load:** Final consumer of AC power.

5.3 Working Principle (Step-by-Step)

1. Sunlight (photons) falls on the PV cell's p-n junction semiconductor.
2. Photons transfer energy to electrons, making them free to move.
3. The built-in electric field at p-n junction drives electrons in one direction — generating DC current.
4. Multiple cells form a module; multiple modules form a panel/array.
5. DC power passes through charge controller to battery and/or inverter.
6. Inverter converts DC → AC for use in homes and industry.

5.4 Neat Labeled Diagram

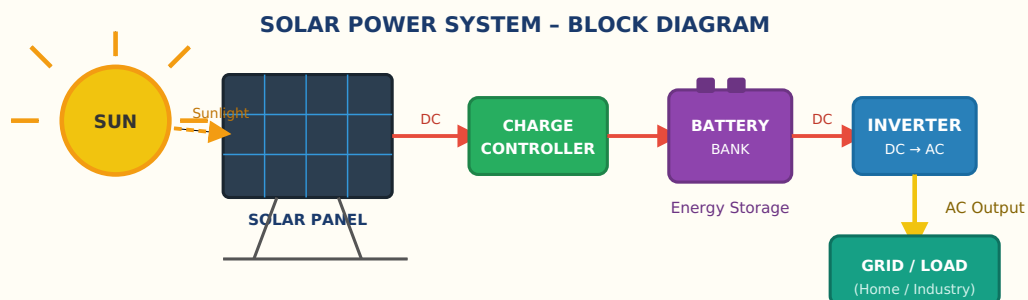


FIG 5.1 - SOLAR POWER SYSTEM BLOCK DIAGRAM

✓ Advantages

- Renewable and inexhaustible energy
- Zero emission / pollution-free
- Low maintenance after installation
- Can be installed in remote areas
- No moving parts (PV cells)

✗ Disadvantages

- No power generation at night
- High initial investment cost
- Depends on weather/cloud cover
- Large area required for high power

5.5 Applications

- Rooftop solar systems for homes (1–10 kW)
- Solar street lights, water pumps, satellites
- Large-scale solar farms (MW to GW scale)

Wind Energy System

6.1 Definition

A **Wind Power System** converts the kinetic energy of wind into mechanical energy using wind turbine blades, which then drives a generator to produce electricity. Wind energy is clean, renewable, and abundantly available.

WIND POWER FORMULA

$$P = \frac{1}{2} \times \rho \times A \times v^3 \times C_p$$

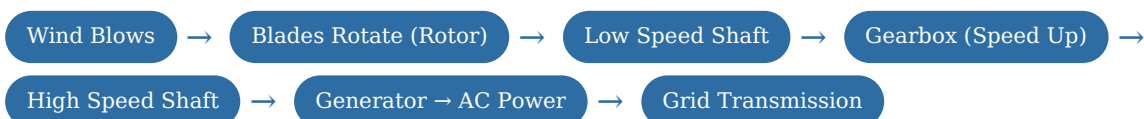
where: ρ = air density (1.225 kg/m^3), A = swept area = πr^2 (m^2), v = wind speed (m/s), C_p = power coefficient (max ≈ 0.59 , Betz limit)

Power is proportional to the cube of wind speed: $P \propto v^3$

6.2 Main Components of a Wind Turbine

1. **Rotor Blades:** Aerodynamically designed blades (usually 3) that capture wind energy and start rotating. Length: 20-80 m.
2. **Hub:** Central part connecting blades to the low-speed shaft.
3. **Nacelle:** Housing on top of the tower containing the gearbox, generator, and controls.
4. **Gearbox:** Converts slow blade rotation (15-20 RPM) to high speed (1200-1800 RPM) for the generator.
5. **Generator:** Converts mechanical rotation to electrical energy (usually AC).
6. **Tower:** Tall structure (50-100 m) supporting the nacelle and rotor at optimal wind height.
7. **Yaw System:** Rotates nacelle to face into the wind direction automatically.
8. **Transformer:** Steps up voltage for grid transmission.

6.3 Working (Step-by-Step)



6.4 Neat Labeled Diagram

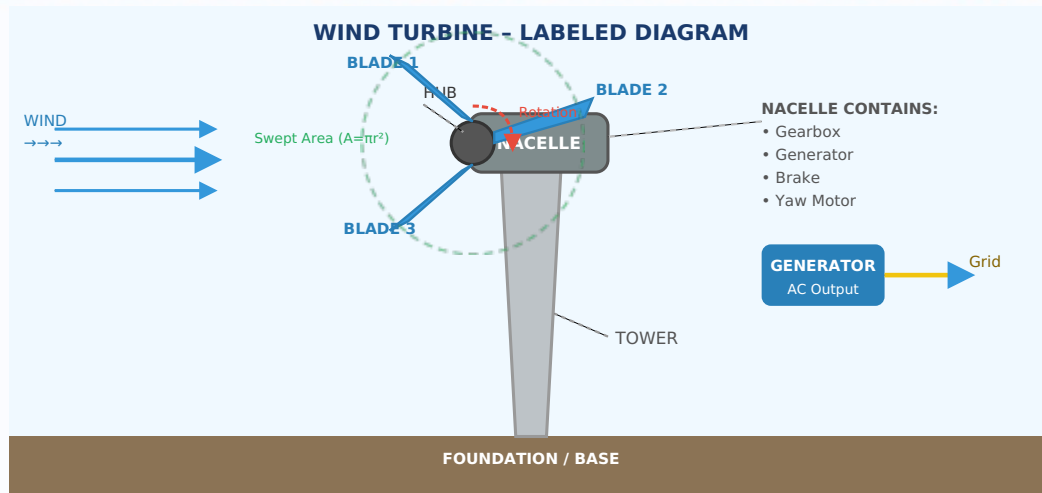


FIG 6.1 - WIND TURBINE (HAWT) LABELED DIAGRAM

✓ Advantages

- Completely renewable and free fuel
- Zero greenhouse gas emissions
- Land below turbines still usable (farms)
- Low operational cost after setup
- Can be installed offshore too

✗ Disadvantages

- Intermittent (depends on wind speed)
- Noise and visual pollution
- Harmful to birds & bats
- Needs minimum wind speed (~3–4 m/s)
- High initial infrastructure cost

Comparison of Energy Sources

Parameter	Thermal	Hydel	Nuclear	Solar	Wind
Fuel / Source	Coal, Oil, Gas	Water	U-235	Sunlight	Wind
Renewable?	No	Yes	No	Yes	Yes
Pollution	High (CO ₂ , SO ₂)	Nil	Radiation risk	Nil	Nil
Initial Cost	Medium	Very High	Very High	Medium-High	Medium-High
Running Cost	High	Low	Medium	Very Low	Very Low
Efficiency	30-40%	80-90%	33-40%	15-22%	35-45%
Load Flexibility	High	High	Low	Low	Low
Land Requirement	Medium	Very Large	Small	Large	Large
Location Dependency	Anywhere	River areas	Anywhere	Sunny areas	Windy areas
Examples (India)	NTPC Ramagundam	Bhakra Nangal	Kudankulam	Gujarat Solar Park	Muppandal, TN

Electricity Bill Concepts

8.1 Unit of Electrical Energy - The kWh

The commercial unit of electrical energy is the **kilowatt-hour (kWh)**, commonly called a "**Unit**". It is the energy consumed by a device rated at 1 kilowatt operating continuously for 1 hour.

ENERGY FORMULAS

Energy (E) = Power (P) × Time (t)

Units (kWh) = [Power in Watts / 1000] × Time in Hours

$E \text{ (kWh)} = (P \times t) / 1000$ [where P is in Watts, t is in hours]

Electricity Bill = Total Units (kWh) × Rate per Unit (₹/kWh)

8.2 Two-Part Tariff

The **Two-Part Tariff** is a method of billing electricity consumers that consists of two components:

Part 1 - Fixed Charge (Demand Charge): A fixed monthly fee based on the consumer's maximum demand (kVA or kW), payable regardless of actual consumption. This covers the capital cost of lines, meters, and infrastructure.

Part 2 - Variable Charge (Energy Charge): Charged based on actual energy consumed (kWh) at a per-unit rate. This covers fuel and running costs.

Total Bill = Fixed Charge + (Energy Consumed × Rate per Unit)

8.3 Types of Tariff

Tariff Type	Description	User Category
Flat Rate Tariff	Fixed rate per kWh regardless of time/load	Small domestic consumers
Block Rate Tariff	Lower rate for higher consumption slabs	Large domestic consumers
Two-Part Tariff	Fixed demand charge + variable energy charge	Industrial/commercial
Time-of-Day Tariff	Different rates for peak/off-peak hours	Large industries
Power Factor Tariff	Penalizes low power factor consumers	Industrial consumers

8.4 Importance of Power Factor

Power factor ($\cos \phi$) represents how effectively electrical power is being used. A low power factor means more current drawn for the same useful power, resulting in higher bills and system losses.

POWER FACTOR & POWER RELATIONS

Active Power $P = V \times I \times \cos \phi$ (Watts or kW)

Apparent Power $S = V \times I$ (VA or kVA)

Reactive Power $Q = V \times I \times \sin \phi$ (VAR or kVAR)

Power Factor = P/S = Active Power / Apparent Power

Ideal Power Factor = 1.0 (unity)

Power Rating of Common Electrical Appliances

Appliance	Typical Power Rating (W)	Typical Usage (hrs/day)	Daily Units (kWh)	Monthly Units (kWh)
Air Conditioner (1.5 ton)	1500	8	12.0	360
Refrigerator	150	24	3.6	108
Washing Machine	500	1	0.5	15
Desktop Computer (PC)	300	8	2.4	72
Laptop	65	8	0.52	15.6
Laser Printer	500	1	0.5	15
LED Bulb (9W)	9	8	0.072	2.16
Tube Light (40W)	40	8	0.32	9.6
Ceiling Fan	75	12	0.9	27
Electric Iron	1000	1	1.0	30
Microwave Oven	900	0.5	0.45	13.5
Water Heater (Geyser)	2000	1	2.0	60
Television (LED 40")	80	6	0.48	14.4
Electric Motor (0.5 HP)	375	4	1.5	45

Note: 1 Unit = 1 kWh = 1000 Wh. Daily Units = (Wattage × Hours) ÷ 1000. Monthly Units = Daily Units × 30 days. These values are approximate and vary by model/usage.

Numerical Problems on Electricity Bill

Problem 1: Basic Electricity Bill Calculation

A household uses the following appliances in a month (30 days):

2 fans (75W each, 12 hrs/day), 5 LED bulbs (9W each, 6 hrs/day), 1 AC (1500W, 8 hrs/day), 1 laptop (65W, 6 hrs/day), 1 TV (80W, 4 hrs/day). Rate: ₹6.50 per unit. Calculate the electricity bill.

Step 1: Energy by fans

$$E_1 = (2 \times 75 \text{ W} \times 12 \text{ h/day} \times 30 \text{ days}) / 1000 = 54,000 \text{ Wh} / 1000 = \mathbf{54 \text{ kWh}}$$

Step 2: Energy by LED bulbs

$$E_2 = (5 \times 9 \text{ W} \times 6 \text{ h/day} \times 30 \text{ days}) / 1000 = 8,100 / 1000 = \mathbf{8.1 \text{ kWh}}$$

Step 3: Energy by Air Conditioner

$$E_3 = (1500 \times 8 \times 30) / 1000 = 360,000 / 1000 = \mathbf{360 \text{ kWh}}$$

Step 4: Energy by Laptop

$$E_4 = (65 \times 6 \times 30) / 1000 = 11,700 / 1000 = \mathbf{11.7 \text{ kWh}}$$

Step 5: Energy by Television

$$E_5 = (80 \times 4 \times 30) / 1000 = 9,600 / 1000 = \mathbf{9.6 \text{ kWh}}$$

Step 6: Total Energy Consumed

$$E_{\text{total}} = 54 + 8.1 + 360 + 11.7 + 9.6 = \mathbf{443.4 \text{ kWh}}$$

Step 7: Electricity Bill

$$\text{Bill} = 443.4 \times ₹6.50 = \mathbf{₹2,882.10}$$

Problem 2: Two-Part Tariff Calculation

An industrial consumer has a maximum demand of 50 kVA. The energy consumed in a month is 8,000 kWh. The tariff is: Fixed charge ₹250 per kVA per month; Energy charge ₹5.50 per kWh. Calculate the total monthly bill and cost per unit.

Step 1: Fixed (Demand) Charge

$$\text{Fixed Charge} = 50 \text{ kVA} \times ₹250 = \mathbf{₹12,500}$$

Step 2: Energy (Variable) Charge

$$\text{Energy Charge} = 8,000 \text{ kWh} \times ₹5.50 = \mathbf{₹44,000}$$

Step 3: Total Bill

$$\text{Total} = ₹12,500 + ₹44,000 = \mathbf{₹56,500}$$

Step 4: Effective Cost per Unit

$$\text{Cost/unit} = \text{Total Bill} / \text{Total Units} = ₹56,500 / 8,000 = \mathbf{₹7.0625 \text{ per kWh}}$$

Problem 3: Energy Savings by Replacing Tube Light with LED

A building has 50 tube lights (40W each, 10 hrs/day). They are replaced with 50 LED lights (9W each). Calculate annual energy savings and cost savings at ₹7/unit.

Old Consumption (Tube Light)

$$E_{\text{old}} = (50 \times 40 \times 10 \times 365) / 1000 = 7,300,000 / 1000 = \mathbf{7,300 \text{ kWh/year}}$$

New Consumption (LED)

$$E_{\text{new}} = (50 \times 9 \times 10 \times 365) / 1000 = 1,642,500 / 1000 = \mathbf{1,642.5 \text{ kWh/year}}$$

Energy Savings

$$\text{Savings} = 7,300 - 1,642.5 = 5,657.5 \text{ kWh/year}$$

Cost Savings

$$\text{Cost Saved} = 5,657.5 \times ₹7 = ₹39,602.50 \text{ per year}$$

Problem 4: Block Rate Tariff

Block Rate Tariff: First 100 units @ ₹3/unit; Next 200 units @ ₹5/unit; Above 300 units @ ₹7/unit. A consumer uses 450 units. Find the monthly bill.

$$\text{First 100 units: } 100 \times ₹3 = ₹300$$

$$\text{Next 200 units (101-300): } 200 \times ₹5 = ₹1,000$$

$$\text{Remaining (301-450) = 150 units: } 150 \times ₹7 = ₹1,050$$

$$\text{Total Bill} = ₹300 + ₹1,000 + ₹1,050 = ₹2,350$$

$$\text{Average cost per unit} = ₹2,350 / 450 = ₹5.22 \text{ per unit}$$

Fuse - Safety Device

11.1 Definition

A **Fuse** is a short piece of thin wire or strip made of metal alloy with a low melting point (tin-lead alloy, aluminum, or silver) that is connected in series with the electrical circuit. It is the simplest and most reliable overcurrent protection device.

11.2 Working Principle

Under normal operating conditions, the fuse wire carries the rated current without heating significantly. When an **overcurrent** (due to short circuit or overload) flows through the circuit, the fuse wire heats up rapidly (I^2R heat), reaches its melting point, and **melts/blows**, thereby breaking the circuit and protecting the equipment from damage.

FUSE FORMULAS

Heat produced $\propto I^2 \times R \times t$ (Joule's Heating Effect)

Fuse Rating (A) must be $>$ Normal Operating Current but $<$ Safe Maximum Current

Preece's Formula: $I = k \times d^{3/2}$ where d = wire diameter (mm), k = material constant

11.3 Neat Labeled Diagram - Fuse

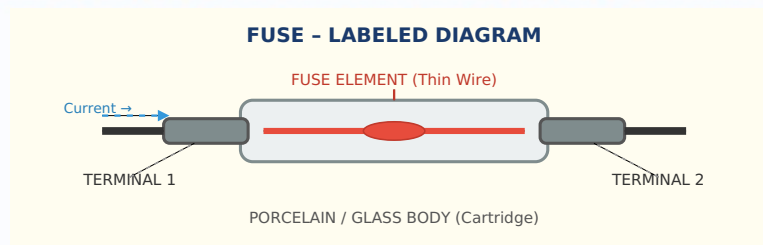


FIG 11.1 - FUSE: LABELED DIAGRAM

11.4 Types of Fuses

- **Rewirable (Kit-Kat) Fuse:** Most common in domestic wiring; fuse wire can be replaced after blowing.
- **Cartridge Fuse:** Sealed unit; better protection; used in industrial/HV systems.
- **High Rupturing Capacity (HRC) Fuse:** Can safely interrupt very large fault currents; used in distribution systems.
- **Miniature Fuse:** Used in electronic equipment and printed circuit boards.

✓ Advantages of Fuse

- Cheapest protective device available
- No separate maintenance required
- Provides automatic protection
- Quickest acting protective device
- Reliable and simple in construction

✗ Disadvantages of Fuse

- Must be replaced after every operation
- Causes supply interruption to all loads
- Exact fuse rating must be used
- Slow to rewire; not suitable for frequent trips
- Cannot distinguish between fault types

11.5 Applications

- Domestic wiring (household circuits)
- Automobile electrical systems
- Electronic equipment protection
- Distribution boards and panel boards

Miniature Circuit Breaker (MCB)

12.1 Definition

A **Miniature Circuit Breaker (MCB)** is an automatically operated electrical switch designed to protect an electrical circuit from damage caused by overload or short circuit. Unlike a fuse, the MCB can be *reset* (*switched ON again*) after it trips — it does not need replacement.

12.2 Working Principle

- An MCB operates on two principles:
- 1. Thermal (Bimetallic) Protection - for Overload:** A bimetallic strip heats up when excess current flows. Due to differential thermal expansion, it bends and trips the mechanism.
 - 2. Electromagnetic Protection - for Short Circuit:** A solenoid (magnetic coil) with a soft iron plunger attracts the plunger instantly when large short-circuit current flows, tripping the MCB in milliseconds.

12.3 Neat Labeled Diagram - MCB

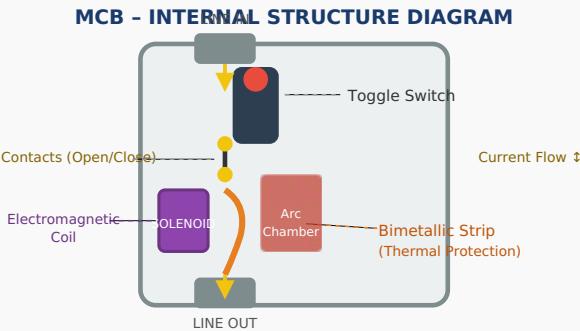


FIG 12.1 - MCB INTERNAL STRUCTURE

12.4 Types of MCB

Type	Tripping Current	Application
Type B	3-5 × rated current	Resistive loads, lighting circuits
Type C	5-10 × rated current	General purpose, small motors
Type D	10-20 × rated current	Heavy industrial motors, welding machines

✓ Advantages of MCB

- Can be reset - no replacement needed
- Dual protection: overload + short circuit
- Trip time can be set accurately
- Safer - no exposed fuse wire
- Easy operation (simple toggle)

✗ Disadvantages of MCB

- Higher initial cost than fuse
- May not trip instantly on very heavy faults
- Mechanical parts may wear out
- Less suitable for very high fault currents (MCCB preferred)

- Does not deteriorate over time

12.5 MCB vs Fuse - Quick Comparison

Parameter	Fuse	MCB
After Tripping	Must be replaced	Reset manually
Cost	Very cheap	Moderate
Speed of operation	Fast	Slightly slower
Protection Type	Overcurrent only	Overcurrent + Short circuit
Indication	Wire broken	Switch trips to OFF
Current setting	Fixed (wire size)	Adjustable (Type B/C/D)

Earthing, Electric Shock & Safety Measures

13.1 Electric Shock

An **electric shock** occurs when an electric current passes through the human body. The body becomes part of the electrical circuit between two points at different potentials. The effects range from a slight tingling sensation to cardiac arrest or death.

Effects of Current on Human Body:

1 mA - Threshold of perception (faint tingling) | 10 mA - Painful; cannot let go | 50 mA - Ventricular fibrillation possible | 100 mA - Fatal if for 1 second | >200 mA - Severe burns, death

SHOCK CURRENT FORMULA

$$I_{\text{shock}} = V_{\text{touch}} / R_{\text{body}}$$

R_{body} (dry skin) $\approx 1 \text{ k}\Omega$ to $100 \text{ k}\Omega$; R_{body} (wet) $\approx 500 \Omega$

With 230V supply: $I = 230 / 500 = 460 \text{ mA} \rightarrow \text{FATAL}$

13.2 Electric Shock Path Diagram

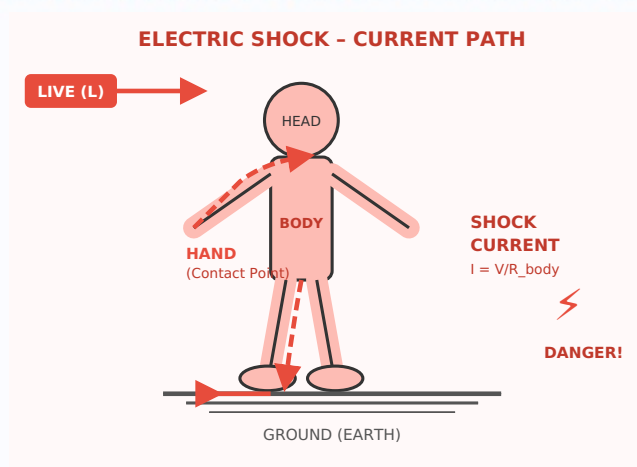


FIG 13.1 - ELECTRIC SHOCK CURRENT PATH THROUGH HUMAN BODY

13.3 Earthing (Grounding)

Earthing (or Grounding) is the process of connecting the metal parts of an electrical system (non-current carrying) to the earth through a low resistance conductor. This ensures that any fault current flows safely into the earth, preventing electric shock to persons touching the equipment.

Purpose of Earthing:

- To protect persons from electric shock in case of insulation failure
- To provide a path for fault current so protective devices (fuse/MCB) operate
- To stabilize voltage levels (reference ground)
- Protection from lightning strikes

13.4 Types of Earthing

Type	Description	Application
Plate Earthing	A copper or galvanized iron plate (60cm×60cm×3mm) buried vertically at 3m depth in moist soil with salt & charcoal layers	Domestic & commercial installations
Pipe Earthing	A GI pipe (38mm dia, 2m long) driven vertically into earth. Most common type in India.	Most widely used; domestic & industrial
Rod Earthing	Copper/steel rod of 12.5mm dia and 2–3m length driven vertically	Rocky or high-resistivity soil
Strip Earthing	Copper strip of 25mm×1.6mm buried horizontally in soil	Large buildings, substations

13.5 Earthing Diagram - Pipe Earthing

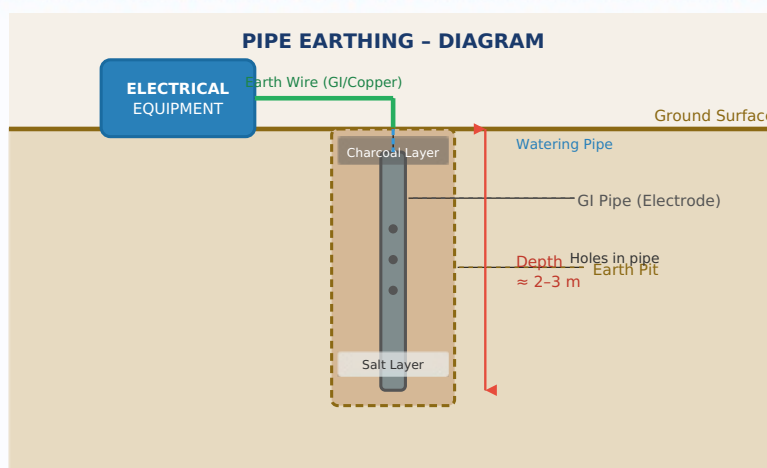


FIG 13.2 - PIPE EARTHING DIAGRAM

13.6 Safety Precautions Against Electric Shock

⚠ MANDATORY SAFETY PRECAUTIONS

1. **Always switch OFF** the mains supply before working on any electrical installation.
2. **Use insulated tools** (rubber/plastic handles) for electrical work.
3. **Never touch** live wires or equipment with bare hands or wet hands.
4. **Proper earthing:** All metal frames and bodies of equipment must be properly earthed.
5. **Use PPE:** Wear rubber gloves, rubber-soled shoes, and safety goggles when working.
6. **Install MCB/Fuse** of correct rating in every circuit.
7. **Use ELCB/RCCB** (Earth Leakage Circuit Breaker) for additional protection.
8. **Avoid overloading** circuits – don't connect too many appliances to one socket.
9. **Inspect wiring** periodically for cracks, breaks, or deterioration of insulation.
10. **Don't use water** on electrical fires – use CO₂ fire extinguisher.
11. **Keep children away** from electrical switchboards and open sockets.
12. **First Aid for shock:** Don't touch victim; switch off power first; call for medical help; give CPR if trained.

13.7 Safety Flowchart

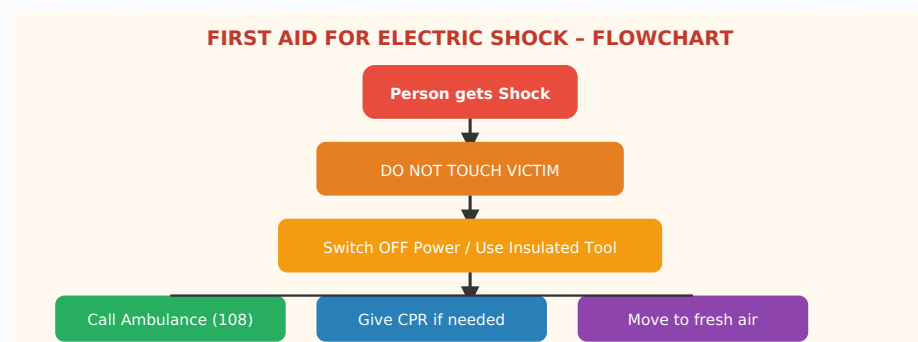


FIG 13.3 - FIRST AID FLOWCHART FOR ELECTRIC SHOCK

13.8 Important Questions & Answers

2-Mark Questions

Q1. Define a fuse. What is its function?

A fuse is a thin wire of low melting point alloy connected in series with a circuit. Its function is to protect the circuit from overcurrent (overload or short circuit) by melting and breaking the circuit when the current exceeds the rated value, thereby preventing damage to equipment and wiring.

Q2. What is earthing? Why is it necessary?

Earthing is the process of connecting metal bodies and non-current-carrying parts of electrical equipment to the earth through low-resistance conductors. It is necessary to provide a safe discharge path for fault/leakage current to earth, preventing electric shock to persons touching faulty equipment and ensuring protective devices (fuse/MCB) operate correctly.

Q3. Define 1 unit (kWh) of electrical energy.

One unit of electrical energy equals 1 kilowatt-hour (kWh). It is the energy consumed when a device of power rating 1 kilowatt operates continuously for 1 hour. $1 \text{ kWh} = 1000 \text{ Wh} = 3.6 \times 10^6 \text{ Joules}$. It is the standard commercial unit for billing electrical energy consumption.

Q4. What is MCB? How does it differ from a fuse?

MCB (Miniature Circuit Breaker) is an automatic switch that protects circuits from overload and short-circuit by tripping. Unlike a fuse, which must be replaced after operation (as the wire melts), an MCB can be simply reset to its ON position after the fault is cleared. MCB provides dual protection (thermal + electromagnetic), is safer, and more convenient than a fuse.

Q5. State the working principle of a solar cell.

A solar cell works on the **photovoltaic effect**. When sunlight (photons) strikes the p-n junction of a semiconductor (Silicon), photons provide energy to electrons. These energized electrons break free from their atoms and create electron-hole pairs. The built-in electric field at the p-n junction drives electrons toward the n-side and holes toward the p-side, creating a potential difference (voltage) and a DC current when an external circuit is connected.

5-Mark / Long Answer Questions

Q1. Explain the working of a Hydel Power Plant with a neat diagram.

Definition: A hydroelectric power plant converts the potential energy of stored water into electrical energy.

Main Components: Dam/Reservoir, Penstock, Surge Tank, Turbine, Generator, Transformer, Tailrace.

Working:

- (1) Rainwater is collected and stored in a dam/reservoir at a considerable height H .
- (2) The stored water possesses potential energy $= mgh$ (where m = mass, g = gravity, h = head).
- (3) Water is released through the penstock (large pipe) and flows downward, converting PE to KE.
- (4) High-velocity water strikes the turbine blades (Pelton/Francis/Kaplan wheel), causing rotation.
- (5) The rotating turbine shaft is coupled to the generator rotor. As rotor spins inside stator windings, AC EMF is induced (Faraday's law).
- (6) The generated AC voltage is stepped up by a transformer for long-distance transmission.
- (7) After passing through the turbine, spent water flows out via the tailrace.

Formula: $P = \rho g Q H \eta$ watts

Refer to Fig 3.1 for the labeled diagram.

Q2. Explain with a neat diagram the working of a Nuclear Power Plant.

Definition: A nuclear power plant generates electricity using heat released from nuclear fission of U-235 atoms.

Principle: When a slow neutron strikes U-235, fission occurs releasing 2-3 fast neutrons and enormous heat. This chain reaction is controlled in a nuclear reactor.

Components and Working:

- (1) *Reactor Core:* Contains U-235 fuel rods arranged in a lattice. Fission occurs here producing heat.
- (2) *Moderator* (Heavy water/graphite) slows neutrons to sustain the chain reaction.
- (3) *Control Rods* (Cadmium/Boron) absorb excess neutrons – pushed in to slow/stop, pulled out to increase reaction.
- (4) *Coolant* (water/heavy water) circulates through reactor, absorbs heat, and carries it to the heat exchanger.
- (5) In the *Steam Generator*, coolant heat converts feed water into high-pressure steam (primary/secondary loop keeps coolant separate).
- (6) Steam drives the *turbine*, which rotates the coupled generator to produce AC electricity.
- (7) The steam is condensed back to water in the *condenser* and recycled.
- (8) Thick concrete and lead *shielding* prevents radiation leakage.

Formula: $E = mc^2$ (Einstein's equation for mass-energy conversion)

Refer to Fig 4.1 for the labeled diagram.

Q3. Explain Two-Part Tariff with a numerical example.

Definition: Two-part tariff is a pricing method for electricity that consists of two components — a fixed demand charge and a variable energy charge.

Components:

- (1) *Fixed Charge (Demand Charge):* Based on maximum demand (kVA or kW), billed monthly regardless of consumption. Recovers fixed infrastructure costs.
- (2) *Variable Charge (Energy Charge):* Based on actual units consumed (kWh). Recovers running/fuel costs.

Total Bill = Fixed Charge + (kWh consumed $\times$ Rate per unit)

Advantages: Fair distribution of costs; encourages efficient use; cost-reflective pricing.

Numerical Example:

Maximum demand = 40 kVA; Energy consumed = 6,000 kWh/month

Fixed charge rate = ₹200/kVA/month; Energy rate = ₹5/kWh

Fixed charge = $40 \times 200 = ₹8,000$

Energy charge = $6,000 \times 5 = ₹30,000$

Total Bill = ₹8,000 + ₹30,000 = ₹38,000

Effective rate = $₹38,000 / 6,000 = ₹6.33$ per unit

Q4. Explain the types of earthing with diagrams.

Earthing connects metal parts of electrical equipment to the earth via a low-resistance conductor to ensure safety.

Types:

(1) *Plate Earthing*: A 60cm×60cm×3mm copper (or GI) plate buried vertically at 3m depth. Packed with layers of charcoal and salt to improve conductivity. Most suitable for high-current applications.

(2) *Pipe Earthing*: A 38mm dia., 2m long GI pipe driven vertically into earth. Salt and charcoal packed around pipe. Most common method in India for domestic and industrial use.

(3) *Rod Earthing*: 12.5mm dia. copper or steel rod of 2-3m length. Used where rocky soil makes plate or pipe installation difficult.

(4) *Strip Earthing*: Copper strip (25mm×1.6mm) buried horizontally in a trench. Used for large installations like substations and transmission towers.

Selection Criteria: Soil resistivity, current magnitude, local conditions, and cost.

Refer to Fig 13.2 for the pipe earthing diagram.

Quick Revision Sheet - Unit III

✦ All Formulas, Key Points & Exam Tips at a Glance

✦ All Important Formulas

Topic	Formula	Units
Hydel Power	$P = \rho g Q H \eta$	Watts (W)
Nuclear Energy	$E = mc^2$	Joules (J)
Wind Power	$P = \frac{1}{2} \rho A v^3 C_p$	Watts (W)
Solar Power	$P = G \times A \times \eta$	Watts (W)
Electrical Energy	$E = P \times t$	Joules or kWh
Units (kWh)	$\text{Units} = (W \times \text{hours}) / 1000$	kWh
Electricity Bill	$\text{Bill} = \text{Units} \times \text{Rate}$	₹ (Rupees)
Active Power	$P = VI \cos \phi$	Watts (W)
Power Factor	$PF = P/S = \cos \phi$	Dimensionless (0-1)
Shock Current	$I = V / R_{\text{body}}$	Amperes (A)
Joule's Heating	$H = I^2 R t$	Joules (J)
Fuse Wire (Preece)	$I = k \times d^{(3/2)}$	A (d in mm)

🔑 Key Points to Remember

- **Thermal Plant:** Fuel → Steam → Turbine → Generator. Uses fossil fuels (coal/oil/gas). High pollution.
- **Hydel Plant:** PE of water → KE → Turbine → Generator. 80-90% efficient. Renewable. Location-dependent.
- **Nuclear Plant:** U-235 fission → Heat → Steam → Turbine → Generator. $E = mc^2$. Radioactive waste issue.
- **Solar PV:** Photovoltaic effect. Sunlight → DC → Inverter → AC. No moving parts. Zero emission.
- **Wind Turbine:** KE of wind → Turbine → Gearbox → Generator. $P \propto v^3$ (cube of wind speed). Betz limit = 59.3%.
- **1 Unit = 1 kWh** = energy consumed by 1 kW device in 1 hour = 3.6×10^6 J
- **Two-Part Tariff:** Total Bill = Fixed Demand Charge + Variable Energy Charge
- **Fuse:** Melts and breaks circuit on overcurrent. Must be replaced after operation. Series connected.
- **MCB:** Trips on overload (thermal) + short circuit (electromagnetic). Can be reset. No replacement needed.
- **Earthing:** Connects equipment body to earth. Types: Plate, Pipe, Rod, Strip. Pipe earthing most

common in India.

- **Electric Shock:** Fatal above 50 mA. Body resistance drops when wet ($\approx 500\Omega$). First aid: Don't touch, cut power, call help.
- **Safety:** Always switch off before work. Use insulated tools. Install ELCB/RCCB. Proper earthing essential.

📐 Diagram Summary - Easy to Draw

Diagram	Key Components to Label	Fig No.
Hydel Power Plant	Reservoir, Dam, Penstock, Turbine, Generator, Tailrace, Transformer, Head (H)	3.1
Nuclear Power Plant	Reactor, Fuel Rods, Control Rods, Moderator, Coolant, Steam Generator, Turbine, Generator, Shielding, Condenser	4.1
Solar Power System	Sun, Solar Panel, Charge Controller, Battery Bank, Inverter, Grid/Load	5.1
Wind Turbine	Blades (3), Hub, Nacelle, Gearbox, Generator, Tower, Foundation, Wind arrows, Swept Area	6.1
Fuse	Terminals (2), Fuse Element/Wire, Cartridge body, Current direction	11.1
MCB	Toggle switch, Contacts, Bimetallic strip, Solenoid, Arc chamber, Line in/out	12.1
Pipe Earthing	Equipment, Earth wire, GI Pipe, Charcoal, Salt, Earth pit, Depth (2-3m)	13.2
Electric Shock Path	Live wire, Human body (head-body-legs), Ground, Current path arrows, Warning	13.1

📌 Exam Tips & Most Important Topics

- ✓ **"Explain with neat diagram"** questions carry 10 marks – always draw labeled diagrams
- ✓ Hydel and Nuclear plant diagrams are asked almost every exam
- ✓ Memorize the wind power formula: $P = \frac{1}{2}\rho A v^3 C_p$ (especially the v^3 part)
- ✓ Two-part tariff numerical – always show fixed charge + energy charge separately
- ✓ MCB vs Fuse comparison table is frequently asked as a 5-mark question
- ✓ Earthing types – pipe earthing is most important; learn all four types
- ✓ Safety precautions – write at least 6-8 points for full marks
- ✓ 1 Unit = 1 kWh must be memorized; use in all bill calculations
- ✓ Electric shock: effects at different current levels (mA) – important for 2-mark
- ✓ Power rating table – AC consumes most energy; LED saves energy

☆☆ End of Unit III - BEEE (JNTUK R23) ☆☆

Best wishes for your examinations! Study smart, draw clear diagrams, and write structured answers.